

Requirement Analysis

Group 1.

Topic: RecipeHub – Recipe Management & Meal Planning Platform.

Business Problem:

Many users store recipes in different places such as notebooks, screenshots, social-media posts, websites, or separate files. Because the recipes are scattered across different sources, users may find it difficult to organize, search, access, and reuse them efficiently.

Planning meals can also become inconvenient when users have to decide what to cook for different days and manually prepare a list of ingredients required for those meals.

RecipeHub aims to address these problems by providing a centralized platform where users can manage and organize their recipes, search and filter recipes based on their requirements, save favorite recipes, plan meals for the week, and generate a shopping list based on the ingredients required for their planned meals.

Business Goal:

The main goal of RecipeHub is to provide a centralized and user-friendly platform for managing recipes and organizing meal planning.

The system aims to allow users to:

- Create, store, and manage their recipes in one place.
- Search and filter recipes according to their requirements.
- Save recipes as favorites for easy access.
- Organize recipes into a weekly meal plan.
- Generate a shopping list based on the ingredients required for the planned meals.

The overall goal is to make recipe management and meal planning more organized, convenient, and efficient for users.

Stakeholders:

Stakeholders are the people or groups who have an interest in the RecipeHub project or are affected by the system.

Stakeholder	Interest / Need
Users	Create, manage, search, save, rate, and plan recipes.
Recipe Creators	Create, publish, and maintain recipes with ingredients and cooking instructions.
Administrators	Manage users, recipe categories, recipes, and system content.
Project Team	Design, develop, test, document, and maintain the system.

Explanation

Users: They are the primary people who use RecipeHub to manage and discover recipes and organize their meal plans.

Recipe Creators: They are responsible for creating and maintaining recipe content, including ingredients and instructions.

Administrators: They are responsible for maintaining and controlling the application, including managing users, categories, and content.

Project Team: The development team is responsible for designing, developing, testing, documenting, and maintaining RecipeHub.

Actors:

Actors are the users or external roles that directly interact with the RecipeHub application.

Actor	Responsibilities
User	Manage personal recipes, search and filter recipes, save favorites, create meal plans, and generate shopping lists.
Administrator	Manage users, recipes, categories, and other administrative content.

Functional Requirements:

Functional requirements describe **what the RecipeHub system should do**. They define the features and operations that users and administrators should be able to perform.

Functional Requirements by Actor:

Actor	Functional Requirements
User	Register and log in; create, view, edit, and delete personal recipes; search and filter recipes; save favorite recipes; rate and review recipes; create weekly meal plans; and view generated shopping lists.
Administrator	Manage users; manage recipe categories; manage inappropriate or invalid recipes and reviews; and view system information.

Core Functional Requirements:

FR-01 — User Authentication

The system shall allow users to **register and log in securely**.

FR-02 — Recipe Creation

The system shall allow an authenticated user to create a recipe containing:

- Recipe title
- Description
- Ingredients
- Cooking instructions
- Preparation time
- Cooking time
- Number of servings
- Difficulty
- Category/Cuisine
- Optional recipe image

FR-03 — Recipe Management

The system shall allow users to **view, edit, and delete recipes** that they are authorized to manage.

FR-04 — Recipe Search

The system shall allow users to **search recipes by name or relevant text**.

FR-05 — Recipe Filtering

The system shall allow users to filter recipes based on:

- Category/Cuisine
- Difficulty
- Cooking time

FR-06 — Favorites

The system shall allow authenticated users to **add or remove recipes from their favorites**.

FR-07 — Ratings and Reviews

The system shall allow authenticated users to **submit ratings and reviews for recipes**, subject to validation.

FR-08 — Weekly Meal Planning

The system shall allow users to **assign recipes to specific days and meal slots in a weekly meal plan**.

FR-09 — Shopping List Generation

The system shall generate a **shopping list based on the ingredients required by the user's planned recipes**.

FR-10 — Administration

The system shall allow administrators to:

- Manage users
- Manage recipe categories
- Manage reported or invalid content

Non-Functional Requirements:

Non-functional requirements describe how the RecipeHub system should behave and the constraints under which it should operate. Unlike functional requirements, they do not describe a specific feature; instead, they define the expected quality and characteristics of the system.

Performance

The system should provide an acceptable response time for common pages and normal user actions under the expected number of users and recipes.

Security

The system shall protect user accounts and user-owned data through proper authentication and authorization.

Usability

The interface should be clear, consistent, and easy to use on common desktop and mobile screen sizes.

Reliability

The system should store valid submitted data consistently and handle errors gracefully.

Availability

The deployed application should be available to authorized users during normal operating periods.

Scalability

The architecture should allow the application to support additional users, recipes, categories, and features without requiring a major redesign.

Maintainability

The application should follow separation of concerns using appropriate components such as **controllers, services, repositories, and data-access components**, making the system easier to understand, test, modify, and maintain.

Functional vs Non-Functional Requirements:

Functional Requirement — What the system does	Non-Functional Requirement — How well / under what constraints
User can create a recipe.	Recipe ownership must be protected through proper authorization.
User can search recipes.	Search results should be provided within an acceptable response time.
User can add a recipe to favorites.	Only authenticated and authorized users can modify their personal favorites.
User can create a weekly meal plan.	Meal-plan changes should be stored reliably.
System can generate a shopping list.	The system should correctly aggregate compatible ingredient quantities where possible.
Administrator can manage users.	Only authorized administrators should have access to administrative functions.

User Stories:

ID	User Story
US-01	As a user, I want to create a recipe so that I can store my recipes in one place.
US-02	As a user, I want to edit or delete my recipes so that I can keep my recipe collection up to date.
US-03	As a user, I want to search and filter recipes so that I can quickly find a suitable recipe.
US-04	As a user, I want to save recipes as favorites so that I can access them easily later.
US-05	As a user, I want to rate and review recipes so that I can record my experience and help other users.

US-06	As a user, I want to create a weekly meal plan so that I can organize what I intend to cook.
US-07	As a user, I want a shopping list generated from my meal plan so that I know which ingredients I need to purchase.
US-08	As an administrator, I want to manage users and recipe categories so that the application remains organized and controlled.

Acceptance Criteria:

Acceptance Criteria — US-01: Create Recipe

User Story: As a user, I want to create a recipe so that I can store my recipes in one place.

No.	Acceptance Criterion
1	The user must be authenticated.
2	The recipe title must be provided.
3	At least one ingredient must be provided.
4	Cooking instructions must be provided.
5	Required values must pass validation.
6	The system must save the recipe with the authenticated user's ownership information.
7	The user must be able to view the newly created recipe.

Acceptance Criteria — US-06: Weekly Meal Plan

User Story: As a user, I want to create a weekly meal plan so that I can organize what I intend to cook.

No.	Acceptance Criterion
1	The user must be authenticated.
2	The user must select a valid day and meal slot.
3	The user must select an available recipe.
4	The meal-plan entry must be saved for the user's account.
5	The user must be able to view and modify the weekly plan.